

PHYSICS 550
Homework #4
Due on Sept 29, 2021

1. G3.1 (5 points)
2. G 3.4 (10 points)
3. G 3.6 (5 points)
- 4.(10 points) let $\hat{A}$ and $\hat{B}$ be Hermitian operators, and define $\hat{C} = [\hat{A}, \hat{B}]$.
 - a Show that $\hat{C}^\dagger = -\hat{C}$. Such an operator is called *anti-Hermitian*.
 - b Show that the eigenvalues of the anti-Hermitian operator $\hat{C}$ are all imaginary.
 - c Suppose $[\hat{K}, \hat{J}] = -s\hat{J}$, where s is a real quantity and $\hat{K}$ is Hermitian.
 - i. Is $\hat{J}$ Hermitian?
 - ii. Show that $[\hat{K}, \hat{J}^\dagger] = +s\hat{J}^\dagger$.
 - iii. Suppose $\hat{K}\varphi_k = k\varphi_k$. Show that $(\hat{J}\varphi_k)$ is an eigenfunction of $\hat{K}$ with eigenvalue $(k - s)$.
 - iv. Show that $(\hat{J}^\dagger\varphi_k)$ is an eigenfunction of $\hat{K}$ with eigenvalue $(k + s)$.
 - v. In the harmonic oscillator, what operators play the role of $\hat{K}$, $\hat{J}$ and $\hat{J}^\dagger$?